

BIOGEOCHEMICAL CYCLES AND ECOLOGICAL PYRAMIDS

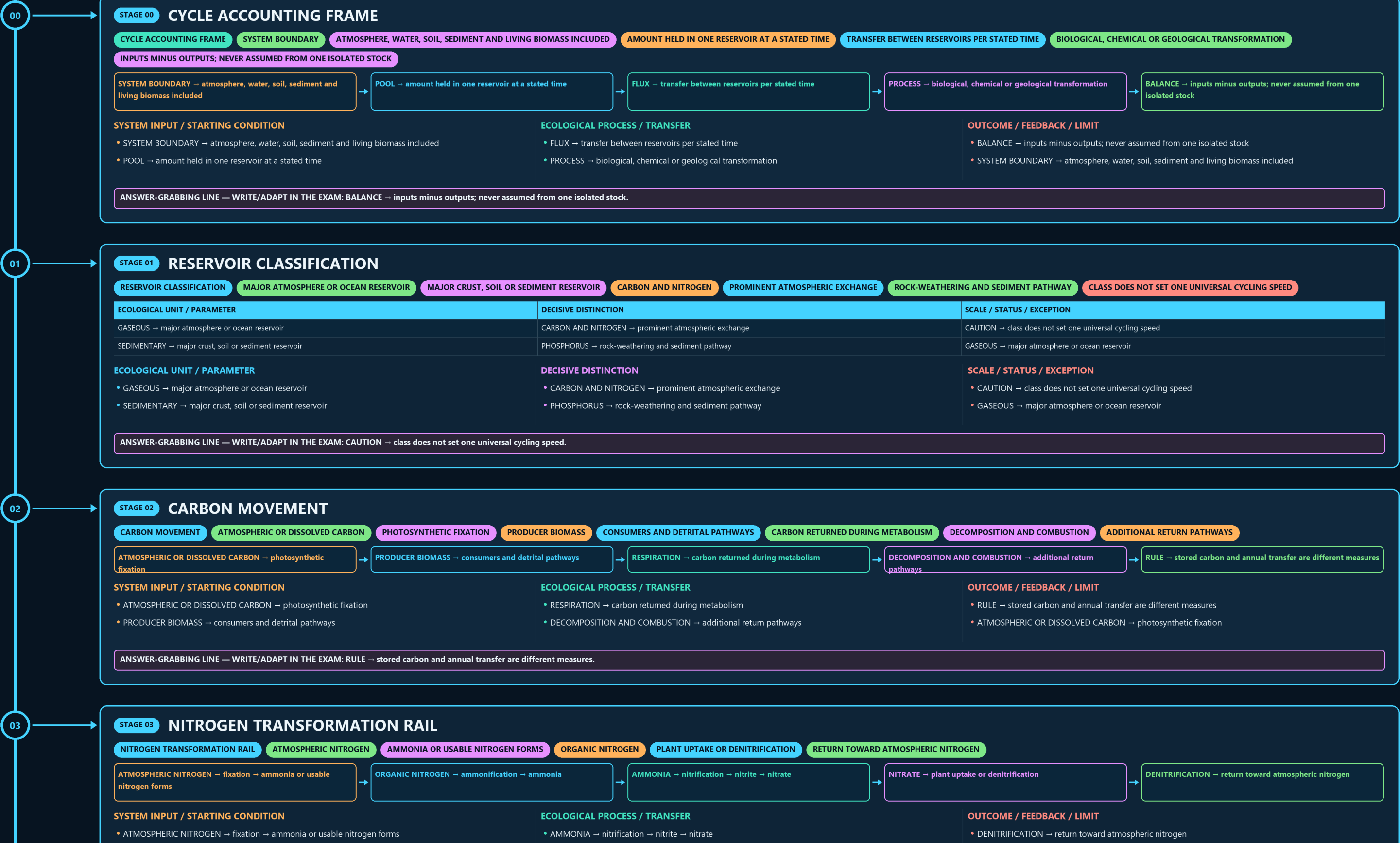
CYCLE ACCOUNTING FRAME → RESERVOIR CLASSIFICATION → CARBON MOVEMENT → NITROGEN TRANSFORMATION RAIL → NITROGEN ORGANISM MAP → PHOSPHORUS PATH

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

• ATMOSPHERIC OR DISSOLVED CARBON → photosynthetic fixation

• PRODUCER BIOMASS → consumers and detrital pathways

• RESPIRATION → carbon returned during metabolism

• DECOMPOSITION AND COMBUSTION → additional return pathways

• RULE → stored carbon and annual transfer are different measures

• ATMOSPHERIC OR DISSOLVED CARBON → photosynthetic fixation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → stored carbon and annual transfer are different measures.

03

STAGE 03

NITROGEN TRANSFORMATION RAIL

NITROGEN TRANSFORMATION RAIL

ATMOSPHERIC NITROGEN

AMMONIA OR USABLE NITROGEN FORMS

ORGANIC NITROGEN

PLANT UPTAKE OR DENITRIFICATION

RETURN TOWARD ATMOSPHERIC NITROGEN

ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms

ORGANIC NITROGEN → ammonification → ammonia

AMMONIA → nitrification → nitrite → nitrate

NITRATE → plant uptake or denitrification

DENITRIFICATION → return toward atmospheric nitrogen

SYSTEM INPUT / STARTING CONDITION

• ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms

• ORGANIC NITROGEN → ammonification → ammonia

ECOLOGICAL PROCESS / TRANSFER

• AMMONIA → nitrification → nitrite → nitrate

• NITRATE → plant uptake or denitrification

OUTCOME / FEEDBACK / LIMIT

• DENITRIFICATION → return toward atmospheric nitrogen

• ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms.

04

STAGE 04

NITROGEN ORGANISM MAP

NITROGEN ORGANISM MAP

SYMBIOTIC FIXATION ASSOCIATED WITH LEGUME ROOTS

FREE-LIVING FIXATION EXAMPLE IN THE OWNER

AMMONIA TO NITRITE STEP

NITRITE TO NITRATE STEP

DENITRIFICATION ROUTE

RHIZOBIUM → symbiotic fixation associated with legume roots

AZOTOBACTER → free-living fixation example in the owner

NITROSOMONAS → ammonia to nitrite step

NITROBACTER → nitrite to nitrate step

SYSTEM / LEVEL / DEFINITION

• RHIZOBIUM → symbiotic fixation associated with legume roots

• AZOTOBACTER → free-living fixation example in the owner

STRUCTURE-FUNCTION MECHANISM

• NITROSOMONAS → ammonia to nitrite step

• NITROBACTER → nitrite to nitrate step

SCALE / EVIDENCE LIMIT

• PSEUDOMONAS-TYPE → denitrification route

• RHIZOBIUM → symbiotic fixation associated with legume roots

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AZOTOBACTER → free-living fixation example in the owner.

05

STAGE 05

PHOSPHORUS PATH

PHOSPHORUS PATH

PHOSPHATE ROCK

SOIL OR WATER PHOSPHATE

PLANT UPTAKE

CONSUMER TRANSFER

DETRITAL RETURN

AQUATIC LOADING

MINING AND FERTILISER

PHOSPHATE ROCK → weathering → soil or water phosphate

PLANT UPTAKE → consumer transfer → detrital return

RUNOFF → aquatic loading → sedimentation

MINING AND FERTILISER → accelerate extraction and dispersal

BOUNDARY → no significant atmospheric phase in the owner

SYSTEM / LEVEL / DEFINITION

• PHOSPHATE ROCK → weathering → soil or water phosphate

• PLANT UPTAKE → consumer transfer → detrital return

STRUCTURE-FUNCTION MECHANISM

• RUNOFF → aquatic loading → sedimentation

• MINING AND FERTILISER → accelerate extraction and dispersal

SCALE / EVIDENCE LIMIT

• BOUNDARY → no significant atmospheric phase in the owner

• PHOSPHATE ROCK → weathering → soil or water phosphate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOUNDARY → no significant atmospheric phase in the owner.

06

STAGE 06

CYCLE-SPECIFIC HUMAN PRESSURE

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FOSSIL-FUEL COMBUSTION AND LAND-COVER CHANGE

SYNTHETIC FERTILISER AND COMBUSTION-RELATED REACTIVE NITROGEN

MINING, FERTILISER APPLICATION AND RUNOFF

CLIMATE FORCING, NUTRIENT POLLUTION AND RESOURCE LOSS DIFFER

SOURCE CONTROL MUST MATCH THE CYCLE AND RESERVOIR

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
CARBON → fossil-fuel combustion and land-cover change	PHOSPHORUS → mining, fertiliser application and runoff	POLICY → source control must match the cycle and reservoir
NITROGEN → synthetic fertiliser and combustion-related reactive nitrogen	OUTCOMES → climate forcing, nutrient pollution and resource loss differ	CARBON → fossil-fuel combustion and land-cover change

ECOLOGICAL UNIT / PARAMETER

• CARBON → fossil-fuel combustion and land-cover change

• NITROGEN → synthetic fertiliser and combustion-related reactive nitrogen

DECISIVE DISTINCTION

• PHOSPHORUS → mining, fertiliser application and runoff

• OUTCOMES → climate forcing, nutrient pollution and resource loss differ

SCALE / STATUS / EXCEPTION

• POLICY → source control must match the cycle and reservoir

• CARBON → fossil-fuel combustion and land-cover change

BIOGEOCHEMICAL CYCLES AND ECOLOGICAL PYRAMIDS • TILE 2/4 • same master rows 2429-5663 of 10522 • continues below

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOUNDARY → no significant atmospheric phase in the owner.

06

STAGE 06 CYCLE-SPECIFIC HUMAN PRESSURE

CYCLE-SPECIFIC HUMAN PRESSURE FOSSIL-FUEL COMBUSTION AND LAND-COVER CHANGE SYNTHETIC FERTILISER AND COMBUSTION-RELATED REACTIVE NITROGEN MINING, FERTILISER APPLICATION AND RUNOFF CLIMATE FORCING, NUTRIENT POLLUTION AND RESOURCE LOSS DIFFER

SOURCE CONTROL MUST MATCH THE CYCLE AND RESERVOIR

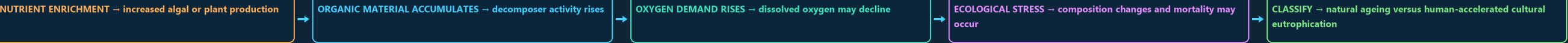
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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: POLICY → source control must match the cycle and reservoir.

07

STAGE 07 EUTROPHICATION MECHANISM

EUTROPHICATION MECHANISM NUTRIENT ENRICHMENT INCREASED ALGAL OR PLANT PRODUCTION ORGANIC MATERIAL ACCUMULATES DECOMPOSER ACTIVITY RISES OXYGEN DEMAND RISES DISSOLVED OXYGEN MAY DECLINE ECOLOGICAL STRESS



SYSTEM INPUT / STARTING CONDITION - NUTRIENT ENRICHMENT → increased algal or plant production - ORGANIC MATERIAL ACCUMULATES → decomposer activity rises	ECOLOGICAL PROCESS / TRANSFER - OXYGEN DEMAND RISES → dissolved oxygen may decline - ECOLOGICAL STRESS → composition changes and mortality may occur	OUTCOME / FEEDBACK / LIMIT - CLASSIFY → natural ageing versus human-accelerated cultural eutrophication - NUTRIENT ENRICHMENT → increased algal or plant production
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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLASSIFY → natural ageing versus human-accelerated cultural eutrophication.

08

STAGE 08 THREE PYRAMID PARAMETERS

THREE PYRAMID PARAMETERS COUNT OF ORGANISMS AT EACH TROPHIC LEVEL STANDING LIVING MATERIAL AT A STATED TIME TRANSFER THROUGH TROPHIC LEVELS OVER TIME DEPENDS ON PARAMETER AND ECOSYSTEM ONLY THE ENERGY PYRAMID IS ALWAYS UPRIGHT

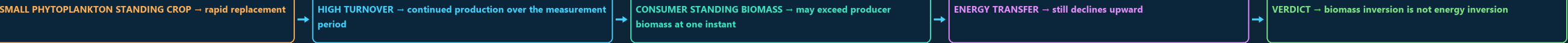
ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
NUMBERS → count of organisms at each trophic level	ENERGY → transfer through trophic levels over time	RULE → only the energy pyramid is always upright
BIOMASS → standing living material at a stated time	SHAPE → depends on parameter and ecosystem	NUMBERS → count of organisms at each trophic level
ECOLOGICAL UNIT / PARAMETER - NUMBERS → count of organisms at each trophic level - BIOMASS → standing living material at a stated time	DECISIVE DISTINCTION - ENERGY → transfer through trophic levels over time - SHAPE → depends on parameter and ecosystem	SCALE / STATUS / EXCEPTION - RULE → only the energy pyramid is always upright - NUMBERS → count of organisms at each trophic level

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → only the energy pyramid is always upright.

09

STAGE 09 AQUATIC BIOMASS INVERSION

AQUATIC BIOMASS INVERSION SMALL PHYTOPLANKTON STANDING CROP RAPID REPLACEMENT HIGH TURNOVER CONTINUED PRODUCTION OVER THE MEASUREMENT PERIOD CONSUMER STANDING BIOMASS MAY EXCEED PRODUCER BIOMASS AT ONE INSTANT ENERGY TRANSFER



SYSTEM INPUT / STARTING CONDITION - SMALL PHYTOPLANKTON STANDING CROP → rapid replacement - HIGH TURNOVER → continued production over the measurement period	ECOLOGICAL PROCESS / TRANSFER - CONSUMER STANDING BIOMASS → may exceed producer biomass at one instant - ENERGY TRANSFER → still declines upward	OUTCOME / FEEDBACK / LIMIT - VERDICT → biomass inversion is not energy inversion - SMALL PHYTOPLANKTON STANDING CROP → rapid replacement
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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → biomass inversion is not energy inversion.

09

STAGE 09

AQUATIC BIOMASS INVERSION

AQUATIC BIOMASS INVERSION

SMALL PHYTOPLANKTON STANDING CROP

RAPID REPLACEMENT

HIGH TURNOVER

CONTINUED PRODUCTION OVER THE MEASUREMENT PERIOD

CONSUMER STANDING BIOMASS

MAY EXCEED PRODUCER BIOMASS AT ONE INSTANT

ENERGY TRANSFER

SMALL PHYTOPLANKTON STANDING CROP → rapid replacement

HIGH TURNOVER → continued production over the measurement period

CONSUMER STANDING BIOMASS → may exceed producer biomass at one instant

ENERGY TRANSFER → still declines upward

VERDICT → biomass inversion is not energy inversion

SYSTEM INPUT / STARTING CONDITION

ECOLOGICAL PROCESS / TRANSFER

OUTCOME / FEEDBACK / LIMIT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → biomass inversion is not energy inversion.

10

STAGE 10

HEURISTIC AND MODEL FIREWALL

HEURISTIC AND MODEL FIREWALL

ENERGY LOSS

QUALITATIVE THERMODYNAMIC DIRECTION IS ROBUST

TEN-PER-CENT RULE

OWNER LABELS IT AN AVERAGE HEURISTIC

WEAKENS A RIGID ONE-LEVEL ASSIGNMENT

CHANGES STANDING STOCKS AND FEEDING RELATIONS

USEFUL SUMMARY, NOT A COMPLETE FOOD WEB

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
ENERGY LOSS → qualitative thermodynamic direction is robust	OMNIVORY → weakens a rigid one-level assignment	PYRAMID → useful summary, not a complete food web
TEN-PER-CENT RULE → owner labels it an average heuristic	SEASONALITY → changes standing stocks and feeding relations	ENERGY LOSS → qualitative thermodynamic direction is robust

ECOLOGICAL UNIT / PARAMETER

DECISIVE DISTINCTION

SCALE / STATUS / EXCEPTION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TEN-PER-CENT RULE → owner labels it an average heuristic.

11

STAGE 11

PYQ AND EVIDENCE BOUNDARY

PYQ AND EVIDENCE BOUNDARY

ROUTED CONCEPTS

AGRICULTURAL NITROGEN, ROCK-WEATHERED PHOSPHORUS, FIXATION LINKS

BASIC PRACTICE

RESERVOIR, PROCESS AND ORGANISM DISTINCTIONS

OFFICIAL PAPER

DEMAND WORDING ONLY

NO KEY

ROUTED CONCEPTS → agricultural nitrogen, rock-weathered phosphorus, fixation links

BASIC PRACTICE → reservoir, process and organism distinctions

OFFICIAL PAPER → demand wording only

NO KEY → no option or answer inferred

LIVE CHECK → no rate, pool, efficiency or status imported from a stub

ECOLOGICAL UNIT / PARAMETER

DECISIVE DISTINCTION

SCALE / STATUS / EXCEPTION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIVE CHECK → no rate, pool, efficiency or status imported from a stub.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

CPCB MONITORS WATER-QUALITY PARAMETERS (BOD, DISSOLVED OXYGEN, NUTRIENT LOAD)

MINISTRY OF AGRICULTURE AND FARMERS WELFARE

DEPARTMENT OF FERTILIZERS

IMPLEMENTATION GAP

FERTILISER SUBSIDY DESIGN IN INDIA HAS HISTORICALLY FAVOURED

CARBON-CYCLE SCIENCE (IPCC AR6, 2023) IS THE SCIENTIFIC SPINE

NATIONAL-SCALE NUTRIENT-CYCLE DISRUPTION (E.G., TOTAL AGRICULTURAL NITROGEN RUNOFF) IS

OPTIONAL ECOLOGICAL PROCESS DEPTH

SCALE, PARAMETER OR STATUS LIMIT

COMPARATIVE RESTORATION NUANCE

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.